AI Based Weather Analyzer

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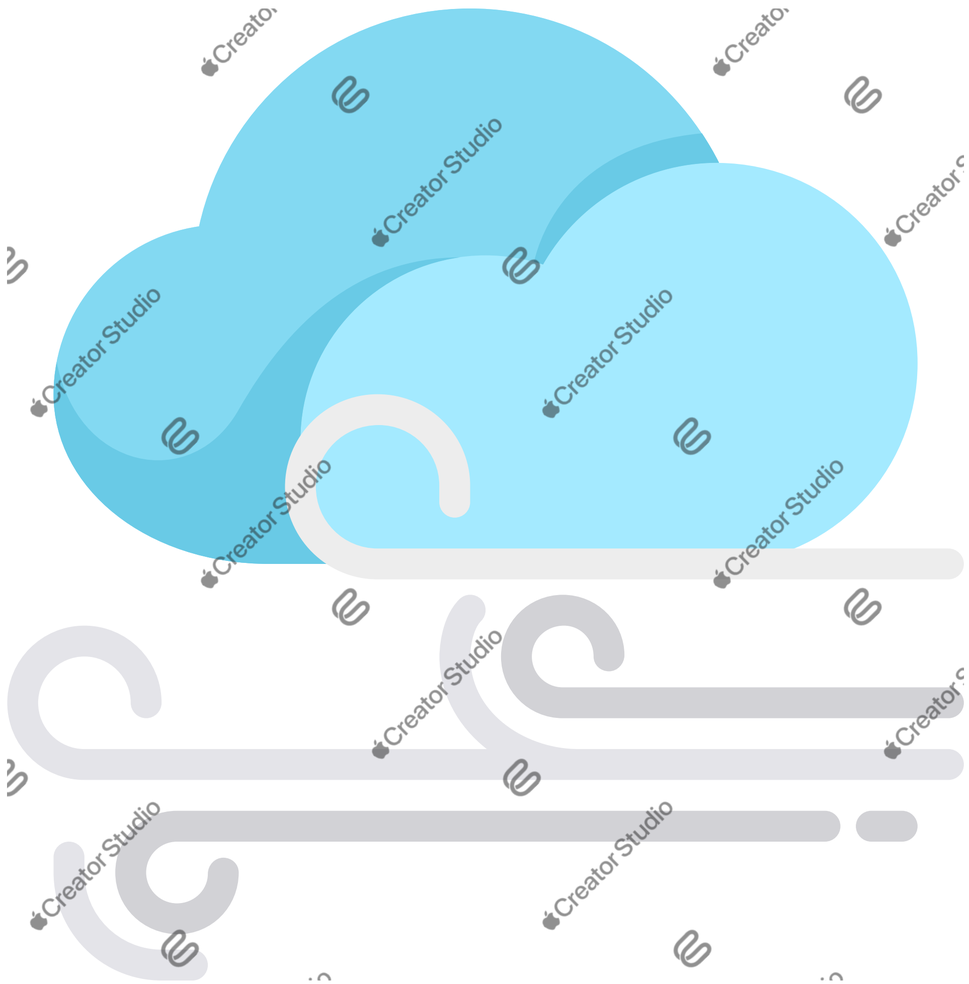


Introduction

Fast Weather Forecasts

The Weather Analyzer is a computer program that uses things like temperature and humidity to make predictions. It can get even better if we add devices that send in information as it happens. We can also use methods together like Random Forest and LSTM to make the predictions more accurate. This is especially helpful when we are trying to figure out what the weather will be like in the future. The Weather Analyzer can tell us if it will be Sunny or Rainy or Cloudy or Hot. This information can be very useful for people who are planning for farms or trying to prepare for storms. The Weather Analyzer is useful for planning and for making sure we are ready, for weather.

Objectives

Goals & Action Plan

- I want to make a system that can tell us what the weather will be like.
- The weather prediction system will use machine learning.
- It will look at things like temperature and humidity to get the weather right.
- The system will use a Decision Tree algorithm to make predictions about the weather.
- I will make a user interface that people can use easily I will use Streamlit to make this interface.
- The weather prediction system will be on a platform so people all over the world can use it.
- I will show how to use intelligence to make the system I will use prompt-based coding to do this.
- The weather prediction system and the artificial intelligence will work together to help people know what the weather will be, like

Stakeholders

This project is beneficial for multiple users:

- **General Public:** To get quick weather predictions
- **Students:** To understand machine learning concepts practically
- **Developers:** To learn ML model integration and deployment
- **Researchers:** To explore data-driven weather analysis
- **Educational Institutions:** For teaching AI/ML applications

Programming Languages, Tools, Software

Language for programming:
Python (main development)
Libraries Used:
Pandas (for handling and preprocessing data)
Scikit-learn (a model for machine learning)
NumPy (for doing math)
Tools:
Code editor: Visual Studio Code
Streamlit is a framework for web applications.
Deployment:
Render (a platform for cloud hosting)

Prompts Used to Generate Code

This project was made with AI-assisted coding and prompts like:

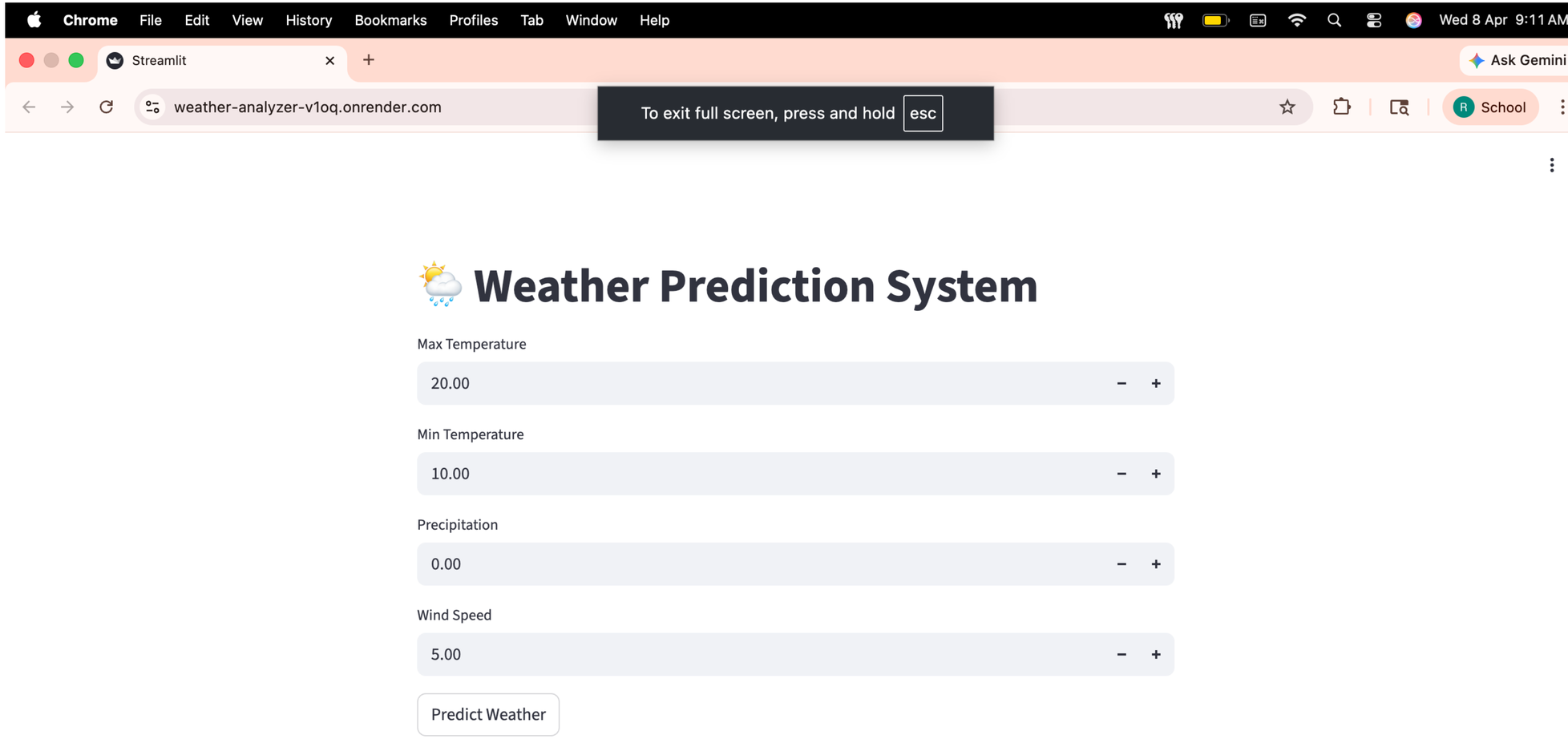
“ Generate python code for weather analyzing for Creating App.py”
“Temperature Analysis “
“Predict Weather “

```
# Generate python code for training a weather prediction model using Decision Tree
Classifier and saving it as model.pkl
# Load dataset
# Features (use correct names from dataset)
# Labels
# Save model
```

Flowchart of Workflow

Dataset → Data Processing → Model Training → Model Saved → User Input → Prediction → Output Display

The dataset is made up of weather data.
Data is loaded and worked on
The model for machine learning is trained.
The model is saved so it can be used again.
User types in values
Model makes guesses about the weather
The output is shown to the user.


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Wed 8 Apr 9:11 AM
Streamlit
Ask Gemini
weather-analyzer-v1oq.onrender.com
To exit full screen, press and hold esc
School
Weather Prediction System
Max Temperature
20.00
Min Temperature
10.00
Precipitation
0.00
Wind Speed
5.00
Predict Weather

Welcome | app.py | requirements.txt | seattle-weather.csv | train_model.py

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```python
# Generate python code for weather analyzing for Creating App.py

import streamlit as st
import pickle

st.title(" Weather Prediction System")

try:
    model = pickle.load(open("model.pkl", "rb"))
except:
    st.error("Model file not found. Please check deployment.")
    st.stop()
# Temperature Analysis
temp_max = st.number_input("Max Temperature", value=20.0)
temp_min = st.number_input("Min Temperature", value=10.0)
precip = st.number_input("Precipitation", value=0.0)
wind = st.number_input("Wind Speed", value=5.0)
# Predict Weather
if st.button("Predict Weather"):
    result = model.predict([[precip, temp_max, temp_min, wind]])
    st.success(f" Weather: {result[0]}")
```

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```python
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
import pickle

# Load dataset
data = pd.read_csv("seattle-weather.csv")

# Features (use correct names from dataset)
X = data[['precipitation', 'temp_max', 'temp_min', 'wind']]

# Labels
y = data['weather']

# Train model
model = DecisionTreeClassifier()
model.fit(X, y)

# Save model
with open("model.pkl", "wb") as f:
    pickle.dump(model, f)

print("Weather model trained successfully!")
```

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References

Official Python Documentation
Scikit-learn Documentation
Documentation for Streamlit
Online guides and lessons
AI tools like ChatGPT for making code and giving advice